Prospective source model of the 2026 M_w 6.8 Kumamoto (Japan) earthquake

A blind test of the MTUQ+WISP pipeline, produced before the USGS finite fault

Automated pipeline (17.Venezuela_2026)

Frozen 2026-07-28, hours after origin — for comparison with the USGS solution when posted

Purpose

No USGS finite-fault model existed for `us6000tgb9` when this was produced. We run the teleseismic pipeline on open GSN/FDSN data and freeze the prediction so it can be compared with the USGS (and JMA/NIED) solutions once published. Only teleseismic body + surface waves and the public W-phase moment tensor (as seed) are used — no strong motion, no GNSS, no InSAR.

1 Event and frozen benchmark

Origin	2026-07-28T07:27:15.5 UTC, 32.682°N, 130.722°E, depth 10 km
Magnitude	M_w 6.8 (W-phase); $M_0 = 1.73 \times 10^{19}$ N m; centroid 11.5 km; DC 87.6 %
Tectonics	shallow crustal strike-slip, Kumamoto fault zone (2016 M_w7.0 sequence region)
W-phase NP1	212.5/75.1/−163.8 right-lateral, NE–SW Futagawa–Hinagu trend
W-phase NP2	118.2/74.3/−15.5 left-lateral conjugate

2 Data

The rewritten fetcher delivered the best coverage of any event in this project: 241 three-component stations (641 channels), 32–89°, maximum azimuthal gap 30°. WISP applied its own quality control on top; MTUQ used response-corrected subsets of 25 and 60 stations.

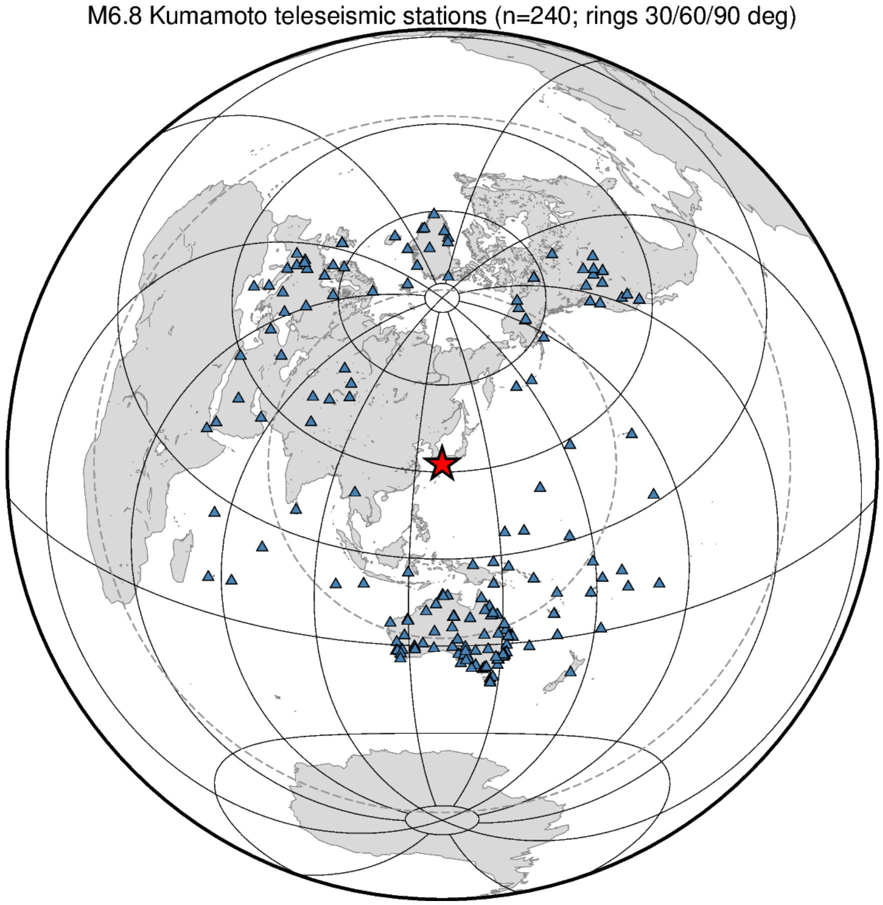

Figure 1: Teleseismic stations (azimuthal-equidistant; rings 30/60/90°).

3 Finite fault (WISP), both nodal planes, **shift_match** applied

WISP `auto_model` on both W-phase planes, seeded from the W-phase moment, followed by the cross-correlation time-shift step and re-inversion (standing rule; it improved the misfit by 9 % and 6 %).

plane	strike/dip/rake	M_w	misfit (pre→post shift)	peak slip	T_{5-95} / V_r
NP2 = Futagawa trend	212/75/−164	6.75	0.1607 → 0.1507	1.88 m	13.6 s / 2.05
NP1 = conjugate	118/74/−16	6.75	0.1667 → 0.1523	1.63 m	14.8 s / 2.09

Table 1: The tectonically expected right-lateral plane fits marginally better (∼1 %). As in our Morocco/Chile/Mexico studies the teleseismic fit is a weak plane discriminant; here the weak preference at least points the same way as the regional tectonics.

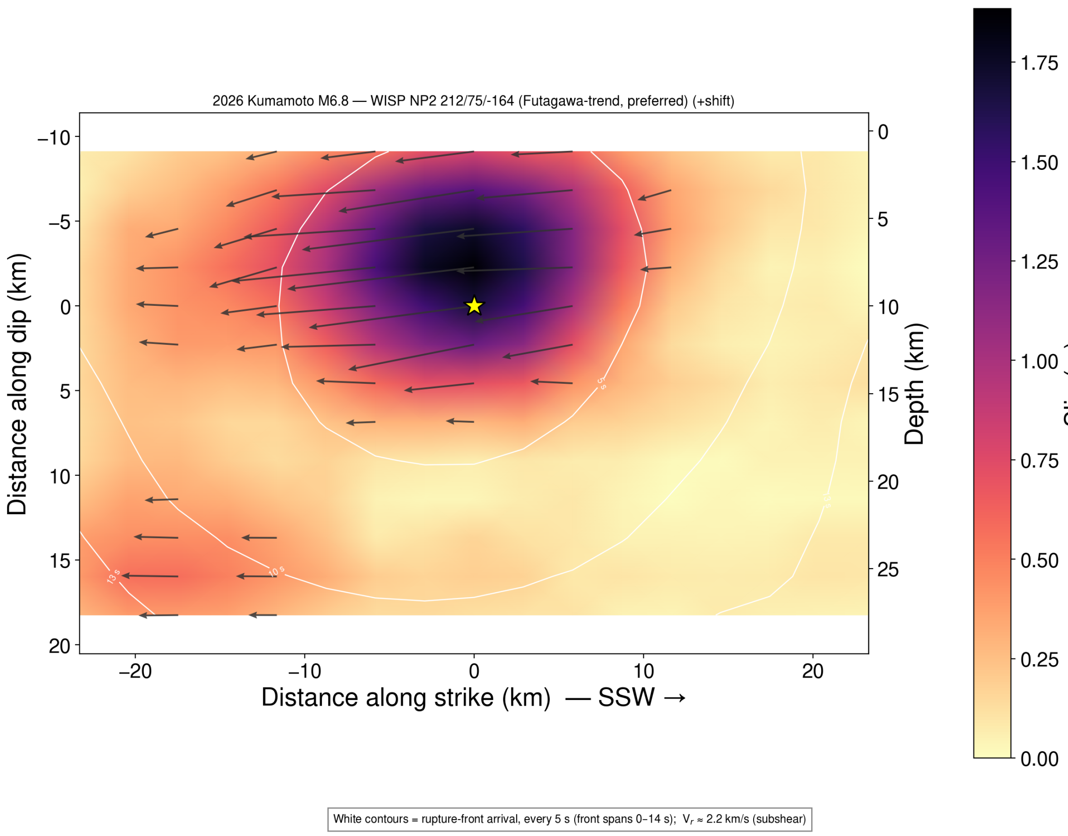

Figure 2: Preferred plane (212/75/−164): single compact asperity slightly updip of the 10-km hypocentre, slip ∼2–16 km depth, peak 1.88 m, rupture front spanning ∼14 s at $V_r \approx 2.1$ km s^{-1} — a smaller sibling of the 2016 Kumamoto rupture.

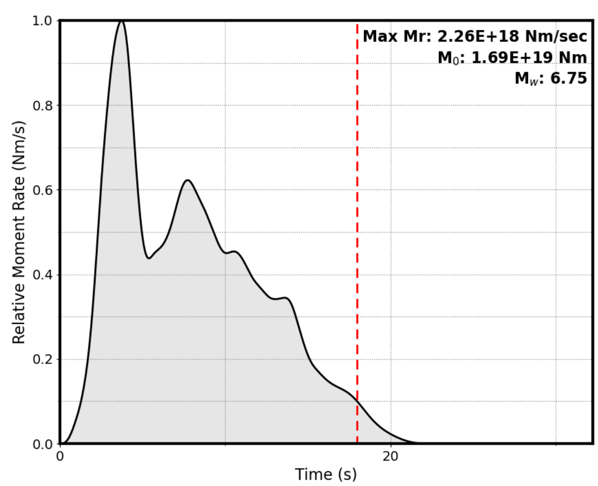

Figure 3: Moment-rate function (preferred plane).

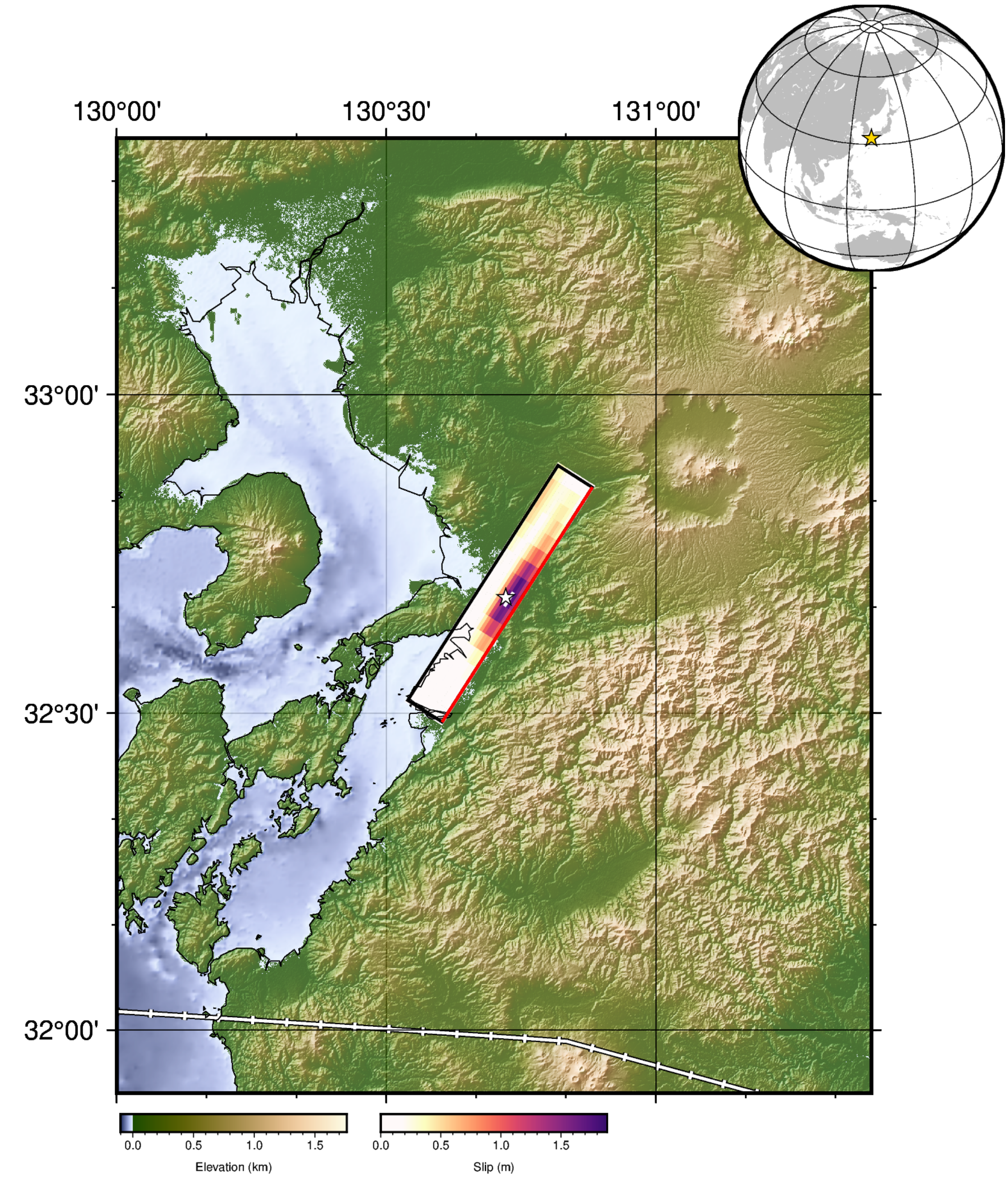

Figure 4: Map view of the preferred-plane slip.

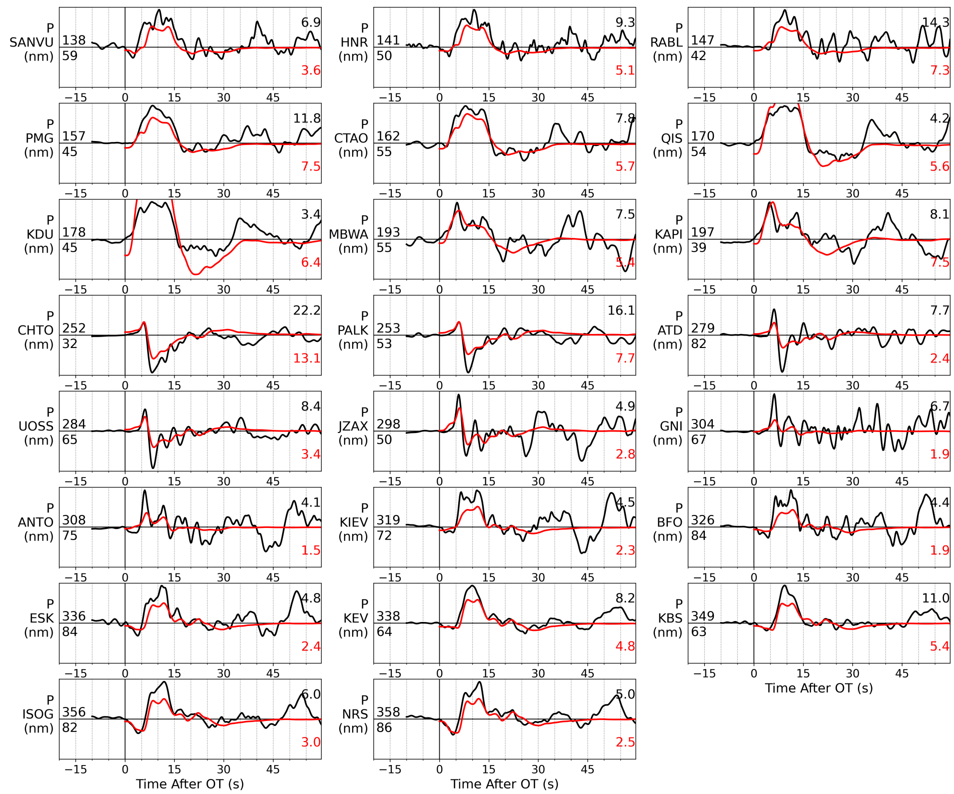

Figure 5: Teleseismic P fits (preferred plane, shifted).

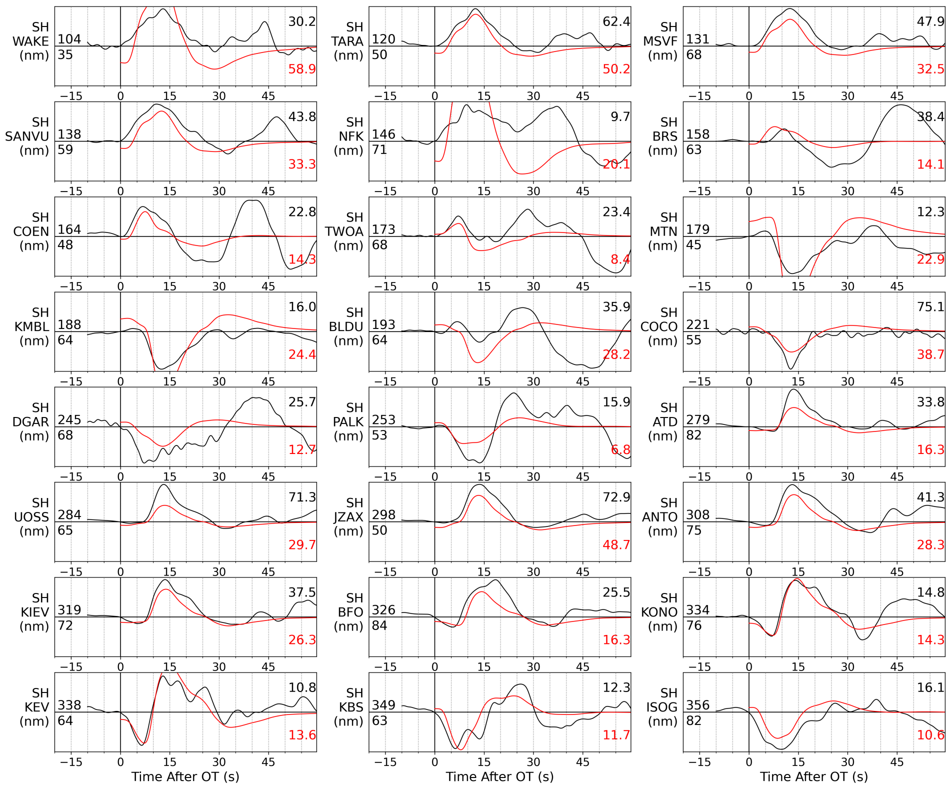

Figure 6: Teleseismic SH fits (preferred plane, shifted).

4 Moment tensor (MTUQ) — an honest failure mode, documented

The independent MTUQ mechanism estimates were unstable for this event, in an instructive way:

variant	stations / gap	mechanism	M_w	Kagan vs W-phase
DC, body P only	25 / 97°	94/59/−29	(edge)	27.1°
DC, body P+SH	25 / 97°	58/50/−83	6.30	67.1°
DC, joint	25 / 97°	184/82/11	6.65	∼78°
full MT (surface)	25 / 97°	198/81/9 (γ=δ=0)	6.60	78.0°
DC, joint, 60 stations	60 / 80°	130/88/7	6.80	27.3°
DC, body P+SH, 60 stations	60 / 80°	212/85/−2	6.75	91.1°

Table 2: MTUQ variants. With the 25-station subset (14 of 25 stations inside one 45° southern azimuth band) the surface-wave solutions lock onto a mechanism rotated ∼90° from the W-phase — the four-lobe aliasing expected for vertical strike-slip radiation, whose patterns repeat every 90° of azimuth. Doubling the stations (gap 97° →80°) snapped the joint solution out of the rotated minimum (78° →27.3°) and recovered M_w 6.80 — confirming the aliasing diagnosis. The body P+SH variant remains aliased; teleseismic SH is unusable at this azimuth geometry. The pipeline's independent mechanism precision here (∼27°) is honestly below the 5–19° achieved for the thrust events.

Two things follow. First, for strike-slip events azimuthal balance matters more than for dip-slip events, because clustered sampling aliases the four-lobed radiation pattern; station count alone is not protection. Second, the shallow-event "body-only" rule from our Morocco/Mexico work is a dip-slip cure (Kanamori–Given indeterminacy of M_{rt}, M_{rp}) and does not transfer to vertical strike-slip, where the long-period surface waves are, in principle, well behaved — given adequate azimuthal sampling.

5 Frozen prospective prediction

- Fault plane: right-lateral 212/75/−164 (Futagawa–Hinagu trend), weakly preferred by the waveform fit and strongly by tectonic context.
- M_w: 6.75 (WISP, both planes), vs W-phase 6.8.
- Slip: single compact asperity, peak ∼1.9 m, ∼2–16 km depth, centred at / slightly updip of the hypocentre; no shallow slip maximum at the surface.
- Duration: $T_{5-95} \approx 14$ s, moment-rate peak at ∼3.5 s; $V_r \approx 2.1$ km s^{-1}.

Caveats. Teleseismic-only; the plane preference is ∼1 % in misfit (weak); the independent point-source mechanism could not be stabilised from the azimuthally clustered long-period subset (documented above), so the mechanism statement leans on the W-phase planes plus the relative WISP fit. USGS will likely add regional/InSAR data; per our Al Haouz study their model may be more compact than ours.